

The Gao–Vitale intrinsic-volume dichotomy is answered negatively

Literature-status note for arXiv:1404.6113

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Status. Exact later-paper answer. This packet records a literature identification and claims no new mathematical result.

Source question

Remark 2.1 on source PDF page 12 of Kabluchko–Zaporozhets [1] repeats Gao and Vitale’s conjecture. If K is a convex GB-compact set in a separable Hilbert space and

$$m_k(K) := (k+1) \frac{V_{k+1}(K)}{V_k(K)},$$

then the proposed dichotomy is

$$\lim_{k \rightarrow \infty} m_k(K) > 0 \quad \text{or} \quad m_k(K) = O(k^{-1/2}).$$

The remark verifies the second alternative for the two Sobolev/Lipschitz examples studied there, but does not settle the universal assertion.

Exact later answer

Dospolova–Germanskov–Zaporozhets [2] restate the identical assertion as Conjecture 3.5 and explicitly disprove it on their PDF page 9. The decisive ingredients are their Theorems 3.1 and 3.4. Theorem 3.1 says that the order $\rho(K)$ of the Steiner entire function of a GB-set can attain every value in $[0, 1]$. Theorem 3.4 proves

$$\limsup_{k \rightarrow \infty} \frac{\log m_k(K)}{\log k} = 1 - \frac{1}{\rho(K)}.$$

Choose any realized $\rho(K) > 2/3$. Then $1 - 1/\rho(K) > -1/2$, so for every $\varepsilon > 0$ the bound

$$m_k(K) = O\left(k^{1-1/\rho(K)-\varepsilon}\right)$$

fails. Taking ε small enough that $1 - 1/\rho(K) - \varepsilon > -1/2$ rules out $m_k = O(k^{-1/2})$. At the same time $\rho(K) < 1$ may be chosen, in which case the displayed limsup is negative and monotonicity of m_k forces $m_k \rightarrow 0$. Thus neither branch of the Gao–Vitale dichotomy holds.

Scope and provenance

The match is exact: arXiv:2507.11626 names the assertion “Conjecture 3.5 (Gao–Vitale (2001))” and explicitly obtains counterexamples by combining its order-realization and decay theorems (in particular, take $2/3 < \rho(K) < 1$). Hence the open-problem signal in arXiv:1404.6113 should be classified as *literature already answered (negative)*, not as a new counterexample produced in this run.

References

- [1] Z. Kabluchko and D. Zaporozhets, *Intrinsic volumes of Sobolev balls with applications to Brownian convex hulls*, arXiv:1404.6113; Remark 2.1, PDF p. 12.
- [2] M. Dospelova, M. Germanskov, and D. Zaporozhets, *On Steiner entire function*, arXiv:2507.11626 (2025); Theorems 3.1 and 3.4, Conjecture 3.5 and its disproof, PDF pp. 7–9.